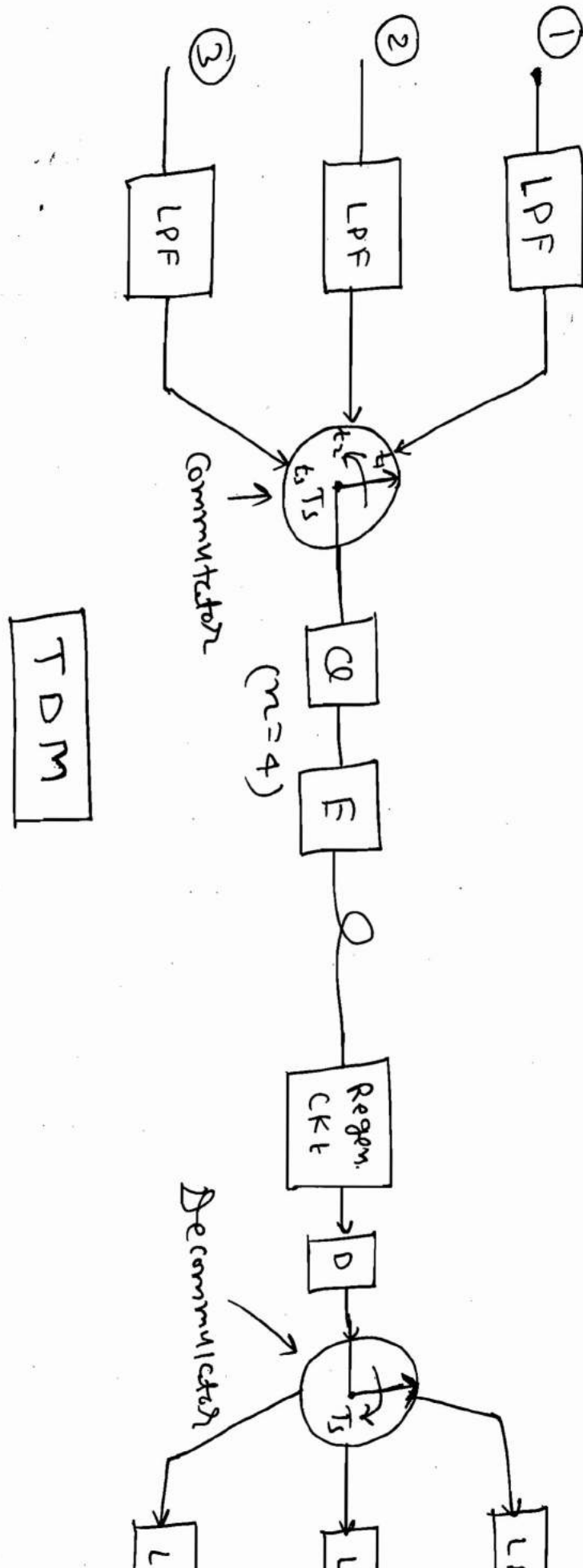




Time

Division

Multiplexing:

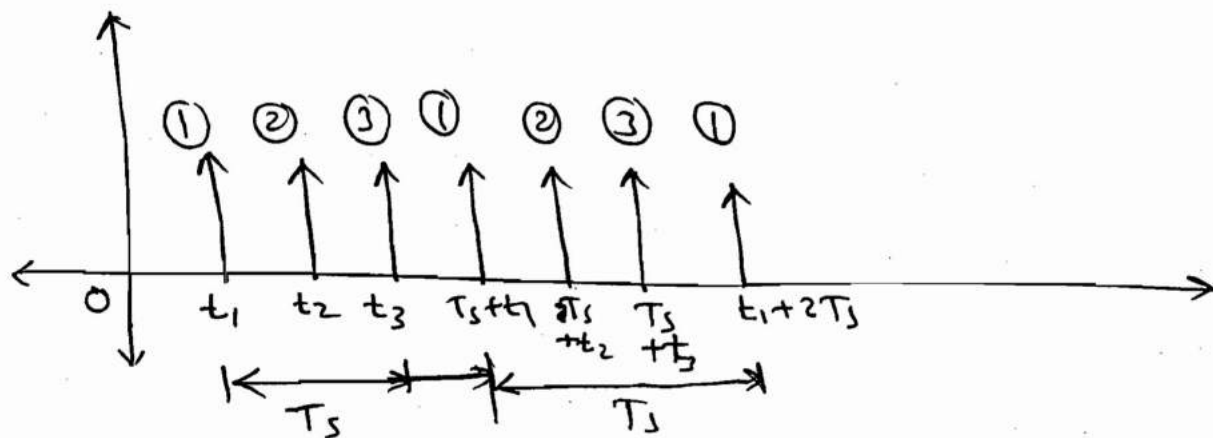


⇒ The LPF is used to eliminate the insignificant high freq.s.

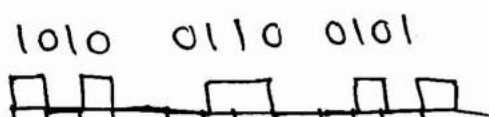
⇒ The Commutator is used to sample the signals for every T_s seconds. The time taken ~~by~~ to complete one revolution of the Commutator is T_s seconds which also indicates the sampling intervals.

⇒ In first revolution of the Commutator the first signal is sampled at t_1 , second signal is sampled at t_2 and third signal is sampled at t_3 .

⇒ The input to the quantizer is shown in figure.



⇒ For every T_s seconds 12 bits are transmitted through the channel,



$$T_b = \frac{T_s}{12} = \frac{T_s}{12}$$

M = no. of Sample (or) no. of signal.

$$\therefore T_b = \frac{T_s}{n \times M}$$

$$\therefore R_b = \left(\frac{1}{T_b} \right) = \frac{n \times M}{T_s}$$

$\therefore$ Bit rate of the
Multiplexed signal,

$$R_b = \frac{1}{T_s} \times n \times M$$

FDM	TDM
$\Rightarrow$ Carrier frequencies are allotted to each signal.	$\Rightarrow$ The time ^{slots} signals are allotted to each signal.
$\Rightarrow$ All the signals are transmitted through the channel at the same time.	$\Rightarrow$ The signals are transmitted in the allotted time only.
$\Rightarrow$ Synchronization is not required.	$\Rightarrow$ Synchronization is required.

Q Five signals each Band limited to 3 KHz are transmitted through a channel using TDM. Each Sample is encoded into 10 bits using PCM. determine the bit rate (or) the multiplexed signals.

Soln: $M=5$, $f_m=3\text{ KHz}$, $n=10$

$\therefore$ Ny. rate = $2f_m = 6000$ Sample/sec.

$\therefore$ sampling rate = $\frac{1}{T_s} = 6000$ sample/sec.

Now, $R_b = \frac{1}{T_s} \times n \times M$

$$R_b = 6000 \times 10 \times 5.$$

$$R_b = 300 \text{ Kbps}$$

Q 10 Voice signals are transmitted through a channel using TDM. Each Sample is encoded into 8 bits. The time taken to complete one revolution of the Commutator is 125 μs . Determine the bit rate of the multiplexed signal.

Ans: $M=10$, $n=8$

$$T_s = 125 \mu\text{s}.$$

$$\therefore \text{Bit rate } R_b = \frac{1}{T_s} \times n \times M.$$

$$\therefore R_b = \frac{10^6}{125} \times 10 \times 8$$

$$= 8000 \times 10 \times 8$$

$$R_b = 640 \text{ kbps}$$



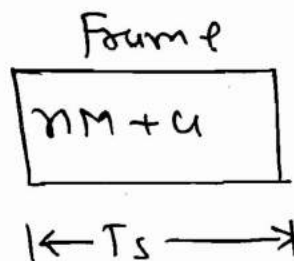
⇒ To Synchronize the Commutator and decommutator the binary data will be transmitted in the form of frames.

The no. of bits generated in one revolution is considered as one frame.

If the duration of the frame is T_s sec.

a frame consist of data bits and synchronization bits.

⇒



$$T_b = \frac{T_s}{12 + 5}$$

$$T_b = \frac{T_s}{nm + a}$$

⇒

$$R_b = \frac{(nm + a)}{T_s}$$

⇒

$$R_b = \frac{1}{T_s} \times (nm + a) \text{ bps.}$$

Q 8 signals are transmitted through a channel using TDM. Each samples is encoded into 10 bits. The speed of the Commutator is 5000 revolution/sec. Determine the bit rate of the multiplexed signal.

- ① If Synchronization requires 5 extra bits per frame.
- ② If Synchronization requires 1 extra bits per sample.

Solⁿ: $m = 8$, $n = 10$, $\frac{1}{T_s} = 5000 \text{ rev/sec}$.

① $a = 5$

$$R_b = \frac{1}{T_s} \times (mn + a)$$
$$= 5000 (80 + 5).$$

$$R_b = 425 \text{ Kbps.}$$

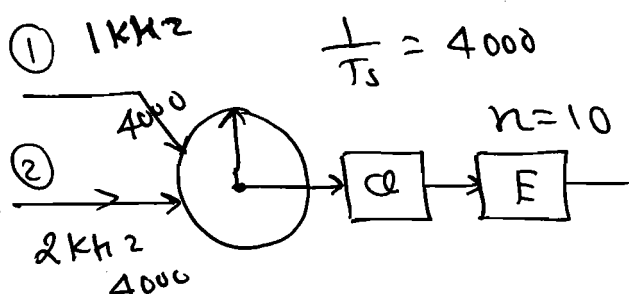
② $a = 1$

$$R_b = \frac{1}{T_s} \times (mn + a).$$
$$= 5000 \times (81).$$

$$\therefore R_b = 405 \text{ Kbps.}$$

*
 → If the freqs are different the following two methods are used.

Method - 1

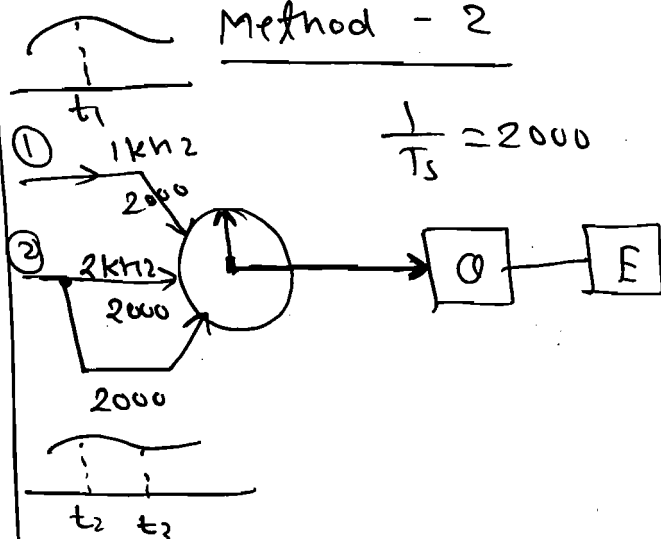


$$R_b = \frac{1}{T_s} \times n \times M$$

$$= 4000 \times 10 \times 2$$

$$R_b = 80 \text{ Kbps}$$

Method - 2



$$R_b = \frac{1}{T_s} \times n \times M$$

$$= 2000 \times 10 \times 3$$

$$R_b = 60 \text{ Kbps}$$

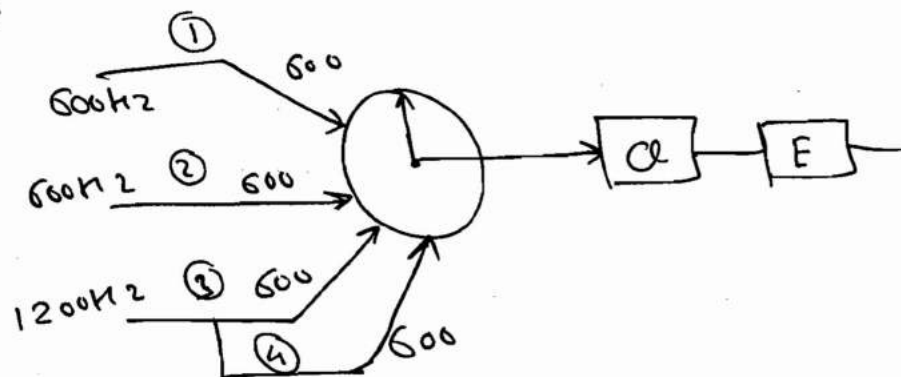
⇒ The bit rate is reduced in second method. As $R_b \downarrow \Rightarrow BW \downarrow \Rightarrow P_e \downarrow$

⇒ The second method is applicable only if the second signal has the multiple integer of the least freq.

☐ Three signals bandlimited to 600 Hz, 600 Hz and 1200 Hz. are sampled at Nyquist rate and transmitted through a channel using TDM. Each sample is encoded

Multiplexed signals.

Soln:



$$\therefore R_b = \frac{1}{T_s} \times n \times M = 1200 \times 12 \times 4 = 57.6 \text{ Kbps.}$$

Q Repeat the above numerical problem if the signals are Band limited to 600 Hz, 1200 Hz & 1800 Hz.

Soln:

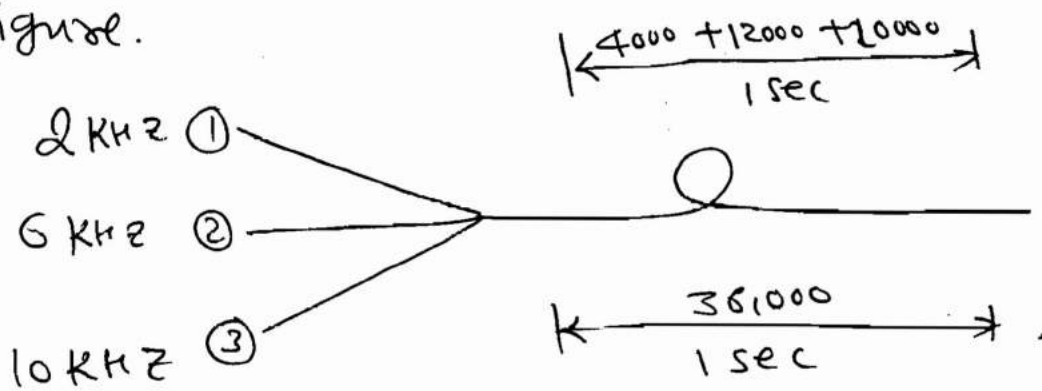
$$R_b = \frac{1}{T_s} \times n \times M$$

$$= 1200 \times 6 \times 12$$

$$\therefore R_b = 86.4 \text{ Kbps.}$$

*
⇒ Consider a signal which is Band limited to 5 KHz and assume the samples are transmitting through a channel. In order to transmit 10,000 samples in 1 sec the minimum BW of channel required is 5 KHz.

⇒ Now, Assume that three signals are multiplexed and samples transmitted through the channel as shown in the figure.

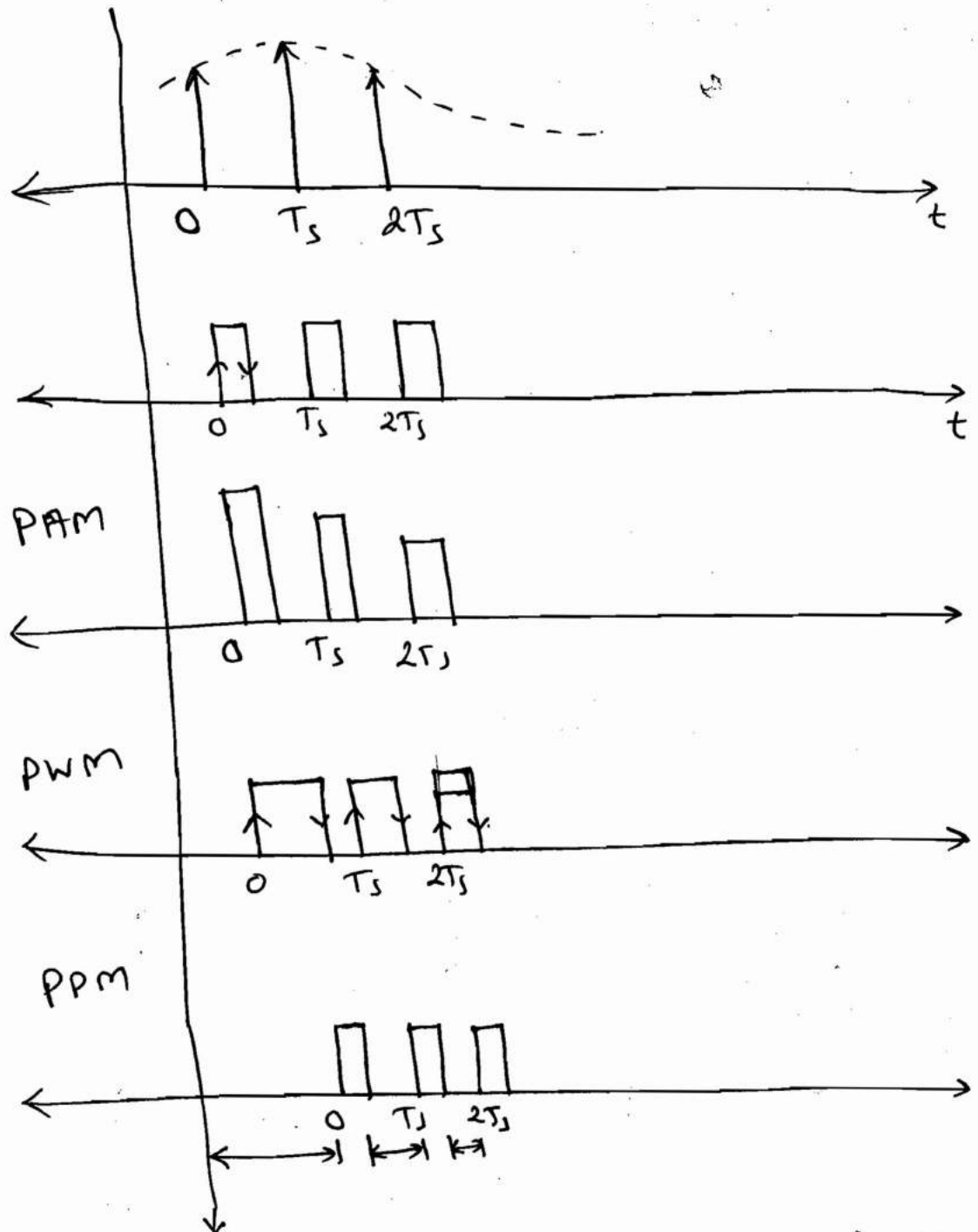


⇒ In the above example, Sampling rate of the multiplexed signals is 36,000 per sec.

⇒ To transmit 36,000 samples in one seconds the minimum BW of the channel required is 18 KHz.

⇒ In Pulse Analog Communication a parameter of the Rectangular pulses is varied according to the sampled value.

⇒ In PAM the power is variable and the BW is constant.



⇒ In PWM power is variable and BW is Var.

⇒ In PPM the power is constant and the

* T₁ Carrier System:-

⇒ T₁ Carrier System is used in telephone exchange to multiplex voice signal using TDM.

⇒ In telephone transmission 24 voice signals are multiplexed to form a T₁ system.

$$\Rightarrow \frac{1}{T_s} = 8000, \quad n = 8$$

$$R_b = \frac{1}{T_s} \times [mn * a]$$

$$R_b = 8000 \times [8 \times 24 + 1]$$

$$R_b = 1.544 \text{ Mbps.}$$

* Digital multiplexer:-

